

The Ground Moves Under the Data

84-355 Democracy's Data | Precinct boundaries, and how to compare two elections

Jonathan Cervas

August 10, 2026

Everything here runs as given. Your job is to read what comes out and say what it means.

Read `precinct-geography-brief.Rmd` first. This lab runs the same steps with the code exposed.

The plan. Every step is done twice: once in **Houston County**, which is small enough to check by hand, and once across Georgia, to show the county is representative rather than convenient. Watch for the places where the two answers differ — those are the interesting ones.

Where did that number come from?

“Turnout in this precinct fell 9 points between 2020 and 2024.”

A sentence like that is in every post-election story, and it contains a hidden claim: **that “this precinct” means the same piece of ground in both years.**

It usually does not. This lab is about what to do instead.

```
# Two helpers used throughout. Precinct keys look like "HOUSTON/ROZR".
cty <- function(x) sub("\\|.*$", "", x)      # county part
nm  <- function(x) sub("^\\|.*", "", x)     # precinct part

# 30 rows of the 2020 shapefile carry no county and no precinct name, so the
# build script collapses them into one pseudo-precinct "NA/NA". It is not a
# precinct. Drop it before any statistic.
ok  <- function(x) !is.na(x) & x != "NA|NA"
junk <- function(d) d[ok(d$from_2020) & ok(d$to_2024), ]
```

Step 1 — Who owns a precinct boundary

Read this, from Georgia's Legislative and Congressional Reapportionment Office, which publishes the statewide precinct map:

“Each individual county across the State of Georgia maintains the official record for the boundaries of their voting precinct lines. County election officials are responsible for any adjustments to these boundaries as well as assigning voters to the specific precinct in which they will vote. The Office of Legislative and Congressional Reapportionment compiles all 159 voting precinct maps provided by the counties into one master shape file...

*This file is a **best fit** version of the maps provided by county election offices, created using current census geography of that decade.”*

Three facts, stated by the state about its own data:

1. **159 counties are the authority.** Not the state.
2. **They adjust boundaries whenever they like.** No cycle, no notice.
3. **The statewide file is a “best fit”** to census geography — which is an admission that precincts do not naturally align to it.

Houston County is one of the 159, and the Houston County Board of Elections is the only body that has ever drawn its precinct lines.

Step 2 — How much moved

```
a <- junk(read.csv(file.path(D, "assign_point.csv"), stringsAsFactors = FALSE))

# statewide
p20 <- unique(a$from_2020); p24 <- unique(a$to_2024)
# one county
h20 <- sort(unique(a$from_2020[cty(a$from_2020) == HC]))
h24 <- sort(unique(a$to_2024[cty(a$to_2024) == HC]))

rbind(
  HOUSTON = c(precincts_2020 = length(h20), precincts_2024 = length(h24),
    in_both = length(intersect(h20, h24)),
    only_2020 = length(setdiff(h20, h24)),
    only_2024 = length(setdiff(h24, h20)),
    pct_surviving = round(100 * length(intersect(h20, h24)) / length(h20), 1)),
  GEORGIA = c(length(p20), length(p24), length(intersect(p20, p24)),
    length(setdiff(p20, p24)), length(setdiff(p24, p20)),
    round(100 * length(intersect(p20, p24)) / length(p20), 1)))
```

```
#>      precincts_2020 precincts_2024 in_both only_2020 only_2024 pct_surviving
#> HOUSTON           16             16      13         3         3          81.2
#> GEORGIA          2651            2539     2240      411       299          84.5
```

Houston has the same *number* of precinct names in both years and they are not the same names: 3 go, 3 arrive. Statewide the survival rate is very close to Houston’s, so this county is a fair sample of the problem, not an outlier.

Measured on the counties’ precinct *codes* rather than their names, statewide survival is only about 75%. **Two keys, two answers, and neither is wrong** — names and codes drift independently, which is the first sign that neither is a stable identifier for a piece of ground.

Either way, a sixth to a quarter of any 2020-to-2024 precinct comparison is comparing a label to itself and nothing else. **Do not fix this by dropping the non-matching precincts** — they are not missing at random.

Every Houston precinct, by name

The county had 16 precincts in 2020, few enough to print in full. This is all of them.

```
aH <- a[cty(a$from_2020) == HC, ]
aH <- aH[order(aH$from_2020), ]
data.frame(precinct_2020 = nm(aH$from_2020),
            became_2024  = nm(aH$to_2024),
            row.names = NULL)
```

```
#>    precinct_2020          became_2024
#> 1          ANNX          ANNX
#> 2           BMS           BMS
#> 3          CENT          CENT
#> 4          CGTC          CGTC
#> 5          FMMS          FMMS
#> 6          HAFS          HAFS
#> 7          HCTC          VFW
#> 8          HEFS          HEFS
#> 9          HHPC          HHPC
#> 10         MCMS          MCMS
#> 11         NSES NORTH HOUSTON SPORT COMPLEX
#> 12         RECR          WELLSTON CENTER
#> 13         ROZR          ROZR
#> 14          TMS          TMS
#> 15         TWPK          TWPK
#> 16          VHS          VHS
```

13 rows say the same thing twice. The other 3 are name changes: HCTC became VFW, NSES became NORTH HOUSTON SPORT COMPLEX, RECR became WELLSTON CENTER. Step 6 will show that in every one of those cases the *ground* did not move at all — only the polling place, and therefore the label.

Then there is a fourth change this table cannot show you, because a precinct-to-precinct table is not able to record a split. Step 5 comes back to it.

Step 3 — The instinct that fails

The obvious repair is to match on precinct name and accept the losses.

```
c(HOUSTON = round(100 * length(intersect(h20, h24)) / length(union(h20, h24))), 1),
   GEORGIA = round(100 * length(intersect(p20, p24)) / length(union(p20, p24))), 1))
```

```
#> HOUSTON GEORGIA
#>    68.4    75.9
```

```
# which Houston precincts does name-matching throw away?
setdiff(nm(h20), nm(h24))
```

```
#> [1] "HCTC" "NSES" "RECR"
```

Matching on name keeps about three quarters of the names in either scope and silently discards the rest. In Houston the discarded precincts are named above, and by Step 6 you will be able to say something sharper about them: **every one of them is a precinct whose boundary did not move**. The rule deletes the places where nothing happened.

Statewide the bias runs the other way as well — growing suburbs split precincts, so fast-changing places also fall out. Either way the discard is not random.

We need a way to put 2020's votes onto 2024's map.

Step 4 — Pick a target, and a common currency

The recipe is always the same:

1. **Choose the newest boundaries as the target.** They are the ones future data will use.
2. **Disaggregate the old data onto a smaller unit that both years contain.**
3. **Reaggregate that unit up to the target.**

The smaller unit here is the **census block** — small enough that precincts rarely cut through one, and nested inside exactly one county.

```
ba <- read.csv(file.path(D, "block_assign.csv"), stringsAsFactors = FALSE)
baH <- ba[!is.na(ba$precinct_2020) & cty(ba$precinct_2020) == HC, ]

rbind(
  HOUSTON = c(blocks = nrow(baH), population = sum(baH$pop, na.rm = TRUE),
    placed_2020 = sum(!is.na(baH$precinct_2020)),
    placed_2024 = sum(!is.na(baH$precinct_2024))),
  GEORGIA = c(nrow(ba), sum(ba$pop, na.rm = TRUE),
    sum(!is.na(ba$precinct_2020)), sum(!is.na(ba$precinct_2024))))
```

```
#>      blocks population placed_2020 placed_2024
#> HOUSTON   3269    163633         3269         3269
#> GEORGIA 232717   10711908        232711        232717
```

Houston's 16 precincts are built out of 3,269 blocks holding 163,633 people, or about 204 blocks per precinct. That is the resolution everything below depends on.

Statewide, six blocks fell outside the 2020 map, almost certainly along the coast where the two agencies drew the shoreline differently. None of them is in Houston. The state population total is Georgia's 2020 census count, to the person.

Step 5 — How a block gets assigned

Not by overlaying polygons. **Reduce each block to a point guaranteed to lie inside it, then ask which precinct contains that point.**

That is the method in `assignPolys` (Cervas, *R-Functions*), and it is more robust than intersecting polygons: no sliver fragments where two agencies drew the same line slightly differently, no fractional pieces, and every block gets exactly one answer.

```
block --interior point--> which 2020 precinct contains it?
                           which 2024 precinct contains it?
```

What the interior point cannot see

The inventory in Step 2 applied the same trick to whole precincts, and that is where its blind spot comes from.

```
h24all <- sort(unique(ba$precinct_2024[!is.na(ba$precinct_2024) &
                                cty(ba$precinct_2024) == HC]))
c(named_by_interior_points = length(h24),
  actually_on_the_2024_map = length(h24all))
```

```
#> named_by_interior_points actually_on_the_2024_map
#>                                16                                17
```

```
setdiff(nm(h24all), nm(h24))
```

```
#> [1] "PEC"
```

PEC is a real 2024 precinct that the interior-point table cannot name. It was carved out of ROZR, and no 2020 precinct has its interior point inside it, so nothing in a precinct-to-precinct table points at it.

That is the whole argument for the block. A precinct reduced to one point can only land in one place, so it can never record a split. Thousands of blocks, each reduced to one point, can.

Blocks are now the common currency. A 2020 precinct's votes can travel to 2024 precincts in proportion to where its people are.

Step 6 — Two ways to weight, and they disagree

```
ar <- junk(read.csv(file.path(D, "crosswalk.csv"), stringsAsFactors = FALSE)) # area
po <- junk(read.csv(file.path(D, "crosswalk_pop.csv"), stringsAsFactors = FALSE)) # people
arH <- ar[cty(ar$from_2020) == HC, ]
poH <- po[cty(po$from_2020) == HC, ]
```

```
split_pct <- function(d, floor = 0) {
  d <- d[d$weight >= floor, ]
  round(100 * mean(table(d$from_2020) > 1), 1)
}
rbind(
  HOUSTON = c(areal_pct_split = split_pct(arH), pop_pct_split = split_pct(poH),
              areal_at_1pct   = split_pct(arH, 0.01),
              pop_at_1pct     = split_pct(poH, 0.01)),
  GEORGIA  = c(split_pct(ar), split_pct(po), split_pct(ar, 0.01), split_pct(po, 0.01)))
```

```
#>      areal_pct_split pop_pct_split areal_at_1pct pop_at_1pct
#> HOUSTON           93.8           6.2           6.2           6.2
#> GEORGIA           78.3           18.5           18.7          14.7
```

Areal weighting says almost every Houston precinct straddles a 2024 boundary. Population weighting says one does. The statewide gap points the same way and is less extreme, so the county sharpens the finding rather than inventing it.

The last two columns say what the gap is made of. Ignore any piece smaller than 1% of a precinct and area's answer collapses onto population's. **Most of what area calls a “split” is a sliver** — a hairline where the 2020 and 2024 files digitised the same road a metre apart, exactly the artefact Step 5 said the interior-point method avoids. What is left over after the slivers are removed is **empty land**: a boundary clipping a corner of farmland with nobody on it.

```
# the one Houston precinct that population weighting agrees was split
po[po$from_2020 == "HOUSTON|ROZR", ]
```

```
#>      from_2020      to_2024  weight  pop
#> 2451 HOUSTON|ROZR HOUSTON|ROZR 0.571018 11908
#> 2452 HOUSTON|ROZR HOUSTON|PEC 0.428982  8946
```

```
ar[ar$from_2020 == "HOUSTON|ROZR", ]
```

```
#>      from_2020      to_2024  weight
#> 5576 HOUSTON|ROZR HOUSTON|PEC 0.555529
#> 5577 HOUSTON|ROZR HOUSTON|ROZR 0.444468
#> 5578 HOUSTON|ROZR HOUSTON|HEFS 0.000002
```

```
# the same split expressed in ballots, from build-houston-figures.R
wm <- read.csv("fig_rozr_meta.csv", stringsAsFactors = FALSE)
wm
```

```
#>  target  w_area  w_pop  pop ballots_area ballots_pop ballots_total
#> 1   ROZR 0.444468 0.571018 11908         4460         5730         10035
#> 2    PEC 0.555529 0.428982  8946         5575         4305         10035
```

ROZR is the only piece of Houston ground that changed precinct in four years. Area puts the line in one place; people put it in another. On the 10,035 ballots ROZR cast in 2020 that gap is 1,270 votes moved from one precinct to another by nothing but a methodological choice.

```
cp <- junk(read.csv(file.path(D, "crosswalk_compare.csv"), stringsAsFactors = FALSE))
cpH <- cp[cty(cp$from_2020) == HC, ]

summarise <- function(d) c(pairs = nrow(d),
                           median_abs_diff = round(median(abs(d$diff)), 3),
                           pct_over_10_points = round(100 * mean(abs(d$diff) > 0.10), 1),
                           worst = round(max(abs(d$diff)), 3))
rbind(HOUSTON = summarise(cpH), GEORGIA = summarise(cp))
```

```
#>      pairs median_abs_diff pct_over_10_points worst
#> HOUSTON    17           0.000           11.8 0.127
#> GEORGIA  3400           0.001            9.5 0.890
```

The rate generalises. The extreme does not. Houston and Georgia disagree by more than ten points of weight at a similar rate — but Houston's worst disagreement is the ROZR split, and Georgia's worst is far uglier:

```
cp[which.max(abs(cp$diff)), ]
```

```
#>               from_2020               to_2024 weight_area
#> 16 CHATHAM|ELI WHITNEY COMPLEX CHATHAM|ELI WHITNEY COMPLEX    0.110056
#>   weight_pop      diff
#> 16          1 0.889944
```

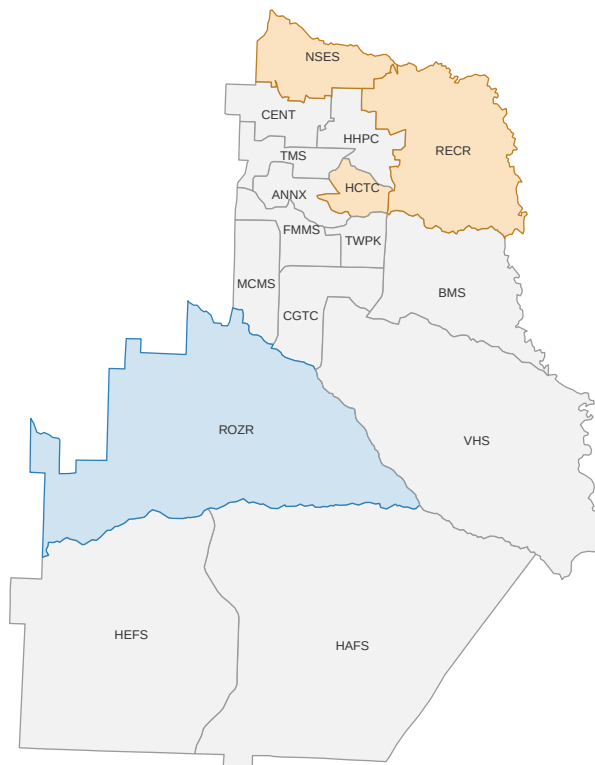
One method sends almost none of that precinct's votes to the target; the other sends nearly all of them. **Nothing like it happens in Houston.** A county is where you learn the mechanism; the state is where you find out how bad it gets.

Areal weighting assumes people are spread evenly across a precinct. They are not. That is the same assumption the redlining lab makes with point-in-polygon, wrong in the same way, and here we can measure how wrong.

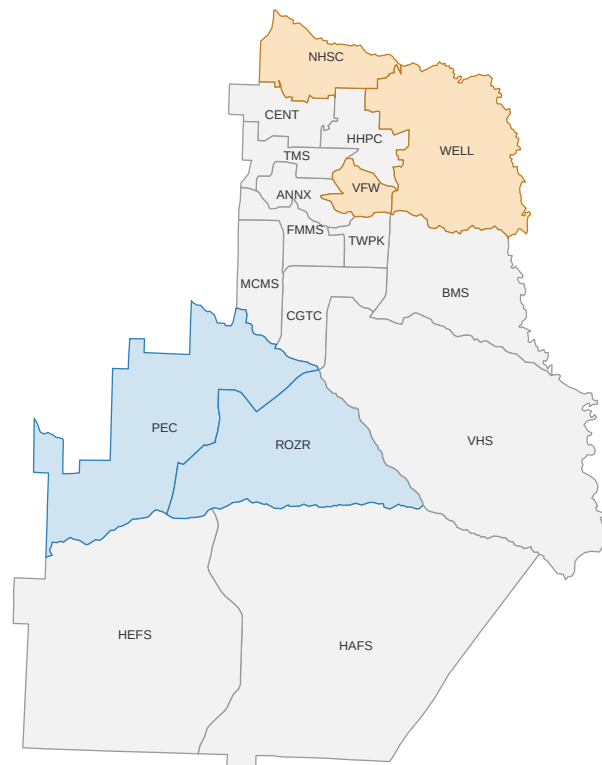
The county, both years

The drawing code for this figure is hidden; it is plumbing, not analysis. The geometry it reads is built by `build-houston-figures.R` in this folder.

2020 – 16 precincts



2024 – 17 precincts



Amber: the name changed and the ground did not — the 4 renames. **Blue:** the ground itself changed — ROZR split, and its western half became the new precinct PEC. **Grey** is unchanged in both name and boundary, which is 12 of 16 precincts.

Step 7 — Carry the votes

```
est <- read.csv(file.path(D, "precincts_2024_pop.csv"), stringsAsFactors = FALSE)
eH <- est[cty(est$to_2024) == HC, ]
eH <- eH[order(-eH$VOTED20), c("to_2024", "TRUMP20", "BIDEN20", "REG20", "VOTED20")]
eH
```

```
#>
#>          to_2024 TRUMP20 BIDEN20 REG20 VOTED20
#> 1962          HOUSTON|BMS      4812      2685 10481      7637
#> 1969          HOUSTON|MCMS      4203      2713  9858      7043
#> 1965          HOUSTON|FMMS      3439      2983  9567      6551
#> 1968          HOUSTON|HHP      2821      3109 10530      6046
#> 1972          HOUSTON|ROZR      3605      2029  8481      5730
#> 1963          HOUSTON|CENT      3130      2353  8366      5584
#> 1977          HOUSTON|WELLSTON CENTER      1087      3878 10539      5054
#> 1976          HOUSTON|VHS      3164      1578  6735      4820
#> 1971          HOUSTON|PEC      2709      1525  6372      4305
#> 1975          HOUSTON|VFW      1909      2022  7184      4021
#> 1964          HOUSTON|CGTC      2236      1374  5182      3680
#> 1973          HOUSTON|TMS      1493      2098  5995      3657
#> 1961          HOUSTON|ANNX      2180      1221  4791      3478
#> 1974          HOUSTON|TWP      1868      1328  4757      3278
#> 1970 HOUSTON|NORTH HOUSTON SPORT COMPLEX      1120      676  2596      1825
#> 1966          HOUSTON|HAF      1083      299  1891      1391
#> 1967          HOUSTON|HEF      675      361  1423      1042
```

```
c(houston_precincts = nrow(eH),
  houston_trump = sum(eH$TRUMP20), houston_biden = sum(eH$BIDEN20),
  georgia_precincts = nrow(est),
  georgia_trump = sum(est$TRUMP20), georgia_biden = sum(est$BIDEN20))
```

```
#> houston_precincts      houston_trump      houston_biden georgia_precincts
#>          17          41534          32232          2692
#>      georgia_trump      georgia_biden
#>      2461823          2474327
```

2020's presidential votes, expressed on 2024's precinct map — including PEC, which did not exist when those votes were cast. Now a precinct-level 2020-to-2024 comparison is a comparison of the same ground.

Step 8 — Check what survived

A crosswalk that loses votes is worse than no crosswalk, because it looks fine.

```
chk <- read.csv(file.path(D, "crosswalk_check.csv"), stringsAsFactors = FALSE)
chk
```

```
#>
#>          check      value
#> 1 2020 precincts (shapefile) 2684
#> 2 carried into the crosswalk 2653
```



```
#> 3 2024 precincts receiving votes      2696
#> 4              votes before 13,821,357
#> 5              votes after 13,821,335
#> 6              votes lost      22
#> 7              share lost (%)      0.000
```

22 votes out of 13,821,357 — 0.000%.

That number was not always this good. An earlier build joined the votes to the shapefile **by precinct name** and lost 489,876 votes — 9.8% of Georgia, concentrated in Chatham and DeKalb, where the two sources spell precincts differently. Two rounds of increasingly clever name-matching moved the loss from 11.1% to 9.8% and then not at all.

The join was never necessary. The precinct shapefile already carries the vote totals in its own attribute table, attached to the geometry. Nothing to match, nothing to lose.

When a join is losing rows, ask whether you needed the join. That is the transferable lesson of this lab, and it cost the person who built it two wrong guesses to learn.

Step 9 — Two state sources, one answer

Was the shapefile's copy of the votes trustworthy? It is published by the reapportionment office; the returns come from the Secretary of State.

```
ts <- read.csv("fig_twosource.csv", stringsAsFactors = FALSE)
ts
```

```
#>      scope matched identical differing shapefile_precincts
#> 1 Georgia    2355      2355         0          2684
#> 2 HOUSTON     16       16         0          16
```

```
round(100 * ts$matched / ts$shapefile_precincts, 1) # what share the join reached
```

```
#> [1]  87.7 100.0
```

On every precinct where the two could be compared, they agreed exactly — 2,355 statewide, and all 16 of Houston's, which is every precinct the county had. Two offices, two publications, the same numbers. That is what licensed using the shapefile's copy and abandoning the name join.

Look at what the join *reached*, though: 87.7% of Georgia's precincts, against 100% of Houston's. **The data was never wrong. The labels were.**

Step 10 — What you can now say

You can say that precinct boundaries are not stable, that 18.8% of Houston's precinct names and 15.5% of Georgia's did not survive four years, and that comparing precincts across years requires putting both on one map rather than matching labels.

You can say that a changed name and a changed boundary are different events. In Houston you can count them separately: 3 renames and 1 real split. Statewide nobody separates them, and the name-survival statistic quietly conflates the two.

You can say how to do it: pick the newest boundaries, disaggregate to census blocks by interior point, reaggregate weighted by population — and check that the votes survive.

You can say that the weighting choice matters, and by how much: statewide it changes 9.5% of the weights by more than ten points, and in Houston it moves 1,270 ballots out of one precinct into another.

You cannot say that population weighting is right. Population is not voters — it counts children and non-citizens, and turnout varies. Registered voters by block would be better, and nobody publishes it.

Your turn

Change **one** thing, re-run, and write about what changed. Pick one.

Option A — Use the areal crosswalk instead. Swap `crosswalk_pop.csv` for `crosswalk.csv` in Step 7 and recompute Houston’s 17 precincts. Which ones move, and can you tell from the map in Step 6 why?

Option B — Find the worst disagreement in Georgia. This one is **statewide, not Houston**: the county’s worst gap is only the ROZR split, and the interesting case is elsewhere. `crosswalk_compare.csv` is sorted by how far the two methods diverge. Take the top row *after* dropping the "NA|NA" pseudo-precinct, look up where it is, and work out what is on that land.

Option C — Turnout, honestly. Using `precincts_2024_pop.csv` and the 2024 returns, compute the change in turnout for Houston’s precincts. Then do it the naive way, by name-matching, and report how different the two answers are — and which 3 precincts the naive method silently drops.

Option D — Your own question. Two maps, and a block-level bridge between them.

What this lab cannot tell you

- **Whether any boundary change was intended to matter.** Counties redraw precincts for polling-place logistics far more often than for anything else, and three of Houston’s four changes were exactly that.
- **Where voters actually live within a block.** The block is the floor here.
- **Anything about 2022.** The state publishes 2012, 2014, 2016, 2018, 2020, 2022 and 2024; this lab uses two of them.
- **Whether population weights are right for turnout.** See Step 10.

On the discussion board

By **Friday, 11:59 p.m.**

- Step 6 shows area and population disagreeing about whether a Houston precinct is “split”. Which question is each answering?
 - Houston lost three precinct names in four years and only one precinct’s ground actually changed. What does that do to a claim built on name survival?
 - Step 8 describes a join that lost 9.8% of the votes and looked fine. How would you have caught it?
 - If a county redraws precincts every cycle, what does that do to the idea of a “bellwether precinct”?
-

Where the data came from

- **Precinct boundaries**, 2020 and 2024 — Georgia Legislative and Congressional Reapportionment Office. <https://www.legis.ga.gov/joint-office/reapportionment> Seven vintages, 2012–2024, all downloadable by script.
- **Census blocks with population** — Census TIGER/Line 2020, Georgia, `t1_2020_13_tabbblock20`.
- **Precinct returns** — Georgia Secretary of State, November 2020.
- **Houston County’s own results**, precinct by precinct, are published at <https://www.houstoncountygga.gov/residents/election-results.cms>. Nothing here uses it. It is listed because it is a genuinely independent third copy of the same totals, so any Houston number in this lab can be checked against it by hand.
- **Assignment method** — Cervas, *R-Functions*, `assignPolys`. <https://github.com/jcervas/R-Functions/tree/main/assignPolys>

Builds are in `../ga-precinct-returns/data/`: `build-crosswalk.R` (areal, plus the interior-point assignment) and `build-block-crosswalk.R` (population, via blocks). The Houston map is built by `build-houston-figures.R` in this folder.